

A comparative evaluation of deep convolutional neural network and deep neural network-based land use/land cover classifications of mining regions using fused multi-sensor satellite data

Ajay Kumar, Amit Kumar Gorai *

Department of Mining Engineering, National Institute of Technology, Rourkela 769008, India

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Abstract

Accurate land use/land cover (LULC) classifications using satellite data is a challenging task due to the limited spatial and spectral resolutions of different satellite data. Moreover, the challenge is more severe for LULC classification of mining regions as no standardized spectral signature is available for the detection of coal mining regions. Thus, the current study aims to design deep learning algorithms using fused data of three satellite sensors (LISS-IV, Landsat-8, and Sentinel-2A) for LULC classification of mining regions. The fused image was derived from three satellite sensors using a discrete cosine transform (DCT) with a spatial correlation approach. A comparative evaluation of deep convolutional neural network (DCNN) and deep neural network (DNN) models in LULC classification of mining regions is conducted. Moreover, the performances of the models with fused data are compared with the performance of the same model with individual sensor data. The study area chosen to execute the work is Jharia Coalfield, which comprises five key LULC types, viz. barren land, coal mining region, built-up area, water body, and vegetation. A total of 6000 image samples of 6×6 sizes and 216,000 pixels were used to train and validate the DCNN and DNN models, respectively. That is, the DCNN model uses the object dataset and DNN uses the pixel dataset for model training and validation. The DCNN model achieved high training and validation accuracies (99.8% and 99.2%), while the DNN model achieved relatively lower accuracies (85.3% and 81.8%). The study evaluates the performance of both models further by employing confusion matrix parameters to measure accuracy, error, precision, and recall for each class. The results reveal that the DCNN model consistently outperforms the DNN model, showcasing accuracy, error rates, precision, and recall ranging from 99.83% to 99.99%, 0.01% to 0.17%, 99.52% to 99.99%, and 99.40% to 99.99% on the training dataset, and 99.50% to 99.99%, 0.01% to 0.50%, 98.35% to 99.99%, and 98.33% to 99.99% on the validation dataset, respectively. In comparison, the DNN model demonstrates values ranging from 90.36% to 99.90%, 0.01% to 9.64%, 75.10% to 99.53%, and 66.99% to 99.99% on the training dataset, and 88.50% to 99.94%, 0.06% to 11.50%, 72.25% to 99.66%, and 62.50% to 99.99% on the validation dataset. These findings showed that the DCNN classification algorithm outperforms the DNN classification algorithm. Moreover, the comparative performances of the DCNN model with different datasets indicate that the model with fused images outperformed the model with individual sensor images.

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1. Introduction

An unscientific mining scheme acts as a threat to the environment, including land-use types (Yu et al. 2018; Chang et al. 2018; Ranjan et al. 2021). It is important to

* Corresponding author.

E-mail address: amit_gorai@yahoo.co.uk (A. Kumar Gorai).

study the mine-induced land use/land cover (LULC) change for sustainable land management (Gao et al. 2021; Chitade and Katyar, 2010; Mborah et al. 2016; Siddiqui and Jain, 2022; Seyam et al. 2023). To achieve this, the remote sensing-based approach can play a crucial role (Chen et al. 2017). In the last few decades, many classification models have been developed by different researchers for LULC studies using satellite or airborne remote sensing data (Ranjan et al. 2021; Kumar and Gorai, 2022a, b; Kumar and Gorai, 2023; Giang et al. 2020; Hufkens et al. 2020; Zhao et al. 2017). It has been observed that the classification accuracy of different algorithms also depends on the spatial resolution of satellite data. The higher spatial resolution offers more accurate delineations of the land-use patterns in comparison to the low spatial resolution data. Moreover, the higher spatial resolution satellite data provides high-quality visualization of objects (Xia et al. 2010). It has been also observed that a limited number of studies on LULC classification using different remote sensing data were conducted for land-use monitoring/detection of mining regions (Ji and Luo, 2021; Madhuanand et al. 2021; Zhou et al. 2023). Chen et al. (2022) designed a joint channel-space attention mechanism and generative adversarial network (JAGAN) for the classification of complex land cover using hyperspectral data. As the land-use types in different mining regions are different, database development for LULC classification using deep learning is also a challenging task. Many researchers/scientists have prepared image datasets from different remote sensing sensors for the development of image classification algorithms (Bodapati & Veeranjanyulu, 2019; Weng et al., 2017; Hamouda et al., 2020; Chen et al., 2016; Tian et al., 2018; Zhang et al., 2019; Sharma et al., 2017; Yang et al., 2018) but the same are not universally applicable and thus a new database generation is required for model development.

Moreover, it is well established that accurate image classification desires high spatial and high spectral resolution data in a single image, which may not be available from a single satellite sensor. Thus, classification using multi-sensor satellite data with fusion provides a crucial starting point for improving classification accuracy (Chen et al. 2017). Currently, many satellite sensors offer spectral values of the earth's surface in a varying set of spectral ranges (bands). Although a satellite sensor has multiple spectral bands, the visualization of the surface characteristics is done using a composite of three bands. Image fusion of different satellite sensors with different spectral and spatial resolutions is a complex process (Kershaw, 1987). The objective of using the fused image for LULC classifications is to improve the classification by enhancing the information content in the input samples. Moreover, the fusion approach assists in enhancing the image quality and offers specific information that is not evident in any individual image, eliminating inaccurate data for improved decision-making. In image fusion, multiple satellite sensors with dif-

ferent spatial resolutions are resampled to a common spatial scale and subsequently, the rescaled images are fused to derive a single image with additional information. In the last few decades, many image fusion techniques have been developed, including weighted arithmetic mean (WAM) (Eltoukhy and Kavusi, 2003), gradient pyramid transforms (GPT) (Petrovic and Xydeas, 2004), morphological pyramid transforms (MPT) (De and Chanda, 2006), Laplacian pyramid transforms (LPT) (Burt and Edward, 1983), discrete wavelet transforms (DWT) (Li et al., 1995), shift-invariant wavelet transforms (SWT) (Rockinger, 1997), and discrete cosine harmonic wavelet transforms (DCHWT) (Kumar, 2013). Out of all techniques, the discrete cosine transform (DCT)-based image fusion technique is very efficient due to low processing time and energy consumption (Haghighat et al., 2011) and adopted in many past studies (Haghighat et al., 2011; Wang and Shang, 2020; Amin-Naji, 2017; Abdipour and Nooshyar, 2016; Cao et al., 2014). In general, the DCT domain divides each reference image into 8×8 blocks and then fuses the reference images into a single output image (Wang and Shang, 2020). In the DCT-based fusion method, different parameters like entropy (Tang, 2004), variance (Haghighat et al., 2011), and frequency (Cao et al., 2014) are used to transform the data for further analysis.

The literature study reveals that many land-use classification algorithms are designed using different datasets, but rarely any study is conducted using multi-sensor dataset for land-use classification of mining regions. Moreover, the existing literature mainly focuses on either DCNN or DNN-based classification methods, but there is limited research available that directly compares their performances using fused images. Also, the performance of the deep learning model with multi-sensor fused image data is rarely done with the performance of the model with individual sensor data. The research gap suggests that there is a need to establish the most effective approach and dataset for the LULC classification of mining regions. Thus, the present study aims to achieve the following objective to fulfill the identified research gaps.

- Generation of a new database from the fused image of three satellite sensors (LISS-IV, Landsat-8, and Sentinel-2A) for LULC classification of mining region using deep learning algorithms. As there is no global LULC database available for coal mining regions, the same can be used for LULC classifications of other coal mining regions. In this study, the classification accuracy obtained from multi-sensor fused data was investigated and compared with the accuracy obtained from individual sensor data.
- A new framework for LULC classification using a deep learning model with a multi-sensor fused dataset is designed.
- The study also investigated the comparative performances of DCNN and DNN models in LULC classifications of mining regions using the fused data.

The manuscript is structured as follows to effectively present the research findings. Section 1 demonstrates the comprehensive background of the study, research gaps, outline of the objectives, and the overall organization of the manuscript. The methodology employed to achieve these objectives is described in detail in Section 2. Also, the description of the key steps of the method is given in Section 2. In Section 3, classification results obtained from the proposed model are demonstrated. Section 4 discusses model development steps and explains the results. Additionally, a comparative evaluation of DCNN and DNN classification algorithms is demonstrated in Section 4. Lastly, the key findings of the work are given in Section 5.

2. Materials and method

2.1. Study area

The study area considered for implementation of the current study is Jharia coalfield (JCF) (Fig. 1). JCF is the oldest coalfield in India, located in the state of Jharkhand. The latitude and longitude of the selected study area (JCF)

are extended from 23.65° N to 23.871° N and 86.313° E to 86.443° E, respectively.

2.2. Method

The research methodology adopted to execute the current study is represented by a flow diagram, as shown in Fig. 2. The current study used three satellite sensors (Landsat8, LISS-IV, and Sentinel 2A) data to derive the fused image, which was subsequently used for the generation of samples for five LULC classes. A total number of 6000 samples (1200 samples for each class) were derived from the fused satellite images. The database for DNN was separately prepared from the extracted image database. All the pixels of the 6000 image samples were used for DNN model training and validation. The object-based sample database and the pixel-based sample database are prepared separately for five LULC classes (barren land, coal mining region, water body, built-up area, and vegetation). The image samples are used for the DCNN classification model, and the point samples are used for classifying the image using the DNN model as shown in Table 1. Finally, the comparative model performances are demonstrated.



Fig. 1. Study area location.

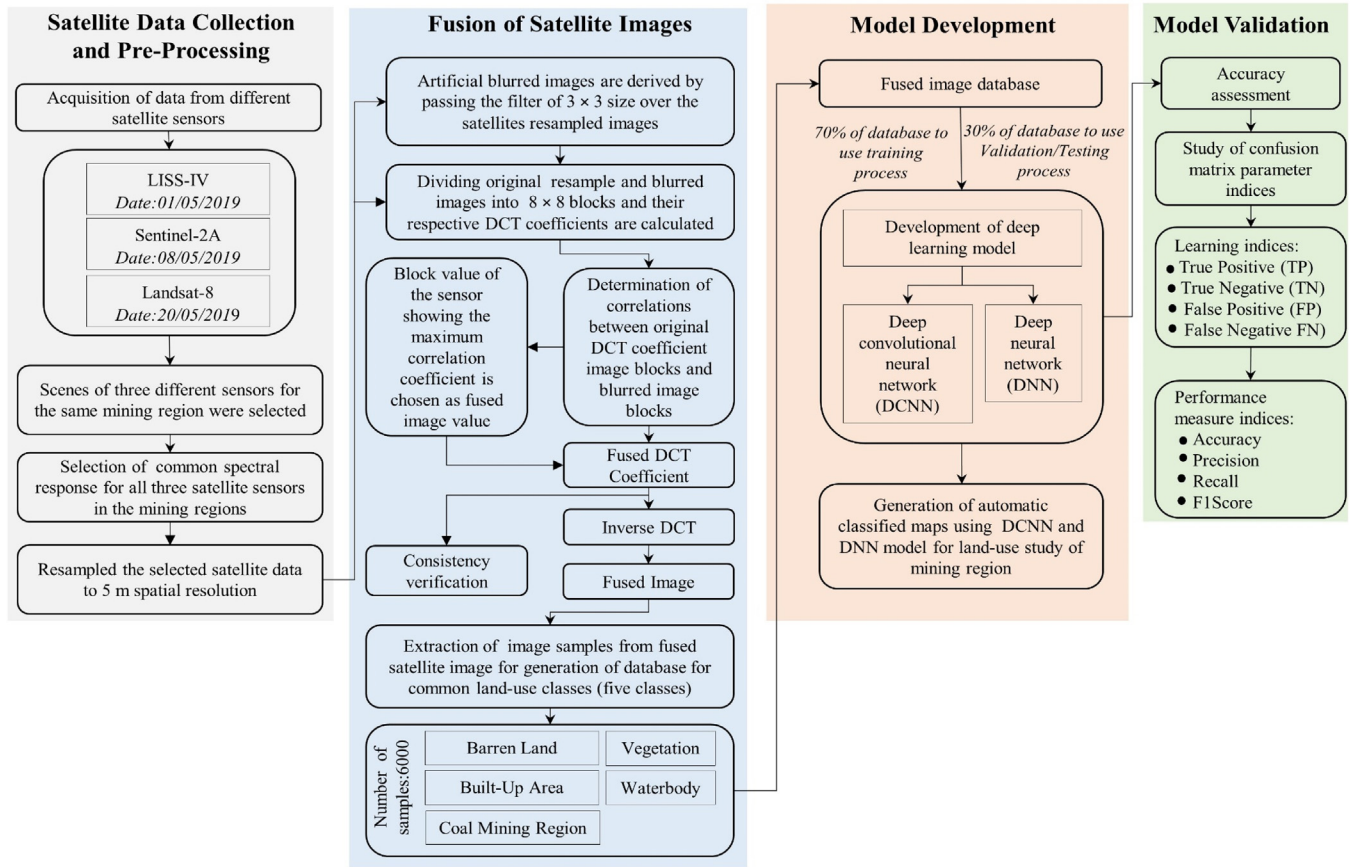


Fig. 2. Flow diagram of adopted research methodology.

Detailed descriptions of the steps are given in the subsequent subsections.

2.2.1. Data acquisition and pre-processing

Landsat-8 and Sentinel-2A satellite scenes captured on 20th May 2019 and 08th May 2019, respectively were downloaded from United States Geological (USGS) Earth Explorer (Source: <https://earthexplorer.usgs.gov/>), and the Linear Imaging Self Scanning Sensor (LISS)-IV scene captured on 1st May 2019 was procured from the National Remote Sensing Centre (NRSC), Hyderabad, India. The satellite data of selected sensors (Landsat-8, Sentinel-2A, and LISS-IV) was chosen for the same period (month and year) depending on the clarity, availability, and season. The satellite data characteristics for selected bands of each sensor are shown in Table 2.

Three bands of all three satellite sensors (LISS-IV, Landsat-8, and Sentinel-2A) with approximately similar spectral profiles in the same wavelength range for five selected LULC types (barren land, coal mining region, water body, built-up area, and vegetation) corresponding were chosen to generate the false colour composite (FCC) images. Landsat-8, LISS-IV, and Sentinel-2A sensors provide the data with their respective spatial resolutions of 30 m, 5.8 m, and 10 m for the selected bands.

2.2.2. Satellite image fusion and image database preparation from fused data

The spatial resolutions of Landsat-8, LISS-IV, and Sentinel-2A satellite sensors are 30 m, 5.8 m, and 10 m, respectively. Each sensor data was resampled to 5 m spatial resolution for the fusion of the three images. After deriving the resampled images with uniform spatial resolution, these are utilized for satellite image fusion. The current study used discrete cosine transform (DCT) with spatial correlation (SC) for the image fusion technique to derive a single image from three satellite sensor images. The fused image preserves all the useful information from the different satellite images into one common image. The framework used in the current study to generate the fused image is shown in Fig. 3. In the first step, the resampled satellite images are fed into the system for fusion. For each image, a corresponding artificial blurred image is derived by passing a low-pass filter of 3×3 size. All the resampled images and their respective blurred images were divided into 8×8 blocks and thereafter the DCT coefficients were computed for each block. It is observed from past studies that block sizes commonly considered for determining the DCT coefficients are 2×2 , 4×4 , 8×8 , and 64×64 . As there is no standardized value exists for consideration of block size, the current study examined the efficiency for all four block

Table 1
Brief outline of a classification system based on deep learning.

SL No	Approach	Classification characteristics	Dataset used	Image input type	Feature extraction	Structural parameters	Hyper-parameters
1	Deep convolutional neural network (DCNN) algorithm	The spatial context and relationships between neighboring pixels (Blaschke, 2003).	Patch-based LULC classes.	Dimensions of the images (width, height, and number of channels)	learnable filters	Input layers with feature maps $[6 \times 6 \times 3]$Three convolutional layers with a number of feature maps 16, 32, and 64 respectivelyTwo max-pooling layers with 2×2 filter size,One fully connected layer with SoftMax function and classification of 5 classes	Learning rate (0.001), Momentum (0.95), Number of epochs (300), Learning rate drop factor (0.1), Batch size (128), and Stochastic gradient descent with momentum optimizer
2	Deep neural network (DNN) algorithm	The spatial context treats each pixel as an independent unit (Zheng et al., 2019).	Pixel-based feature classes.	Pixels of the images (pieces of information)	learnable pixels	Input layers with input sample (108 neurons)Three hidden layers (column/row-wise arrangement of spatial feature value and synapse of information by neural network passes)with 82, 56, and 32 neurons respectivelyOutput layers with classified into five classes of 5 neurons.	Learning rate (0.001), Number of epochs (1000), Validation failure (0 to 6), Gradient (1e-5 to 1e-10), Activation function (sigmoid), and Gradient descent with momentum optimizer

sizes. The efficiency of the fusion process is measured in terms of computational time, signal-to-noise ratio (SNR), and spatial frequency response (SFR), as reported in Table 3. The results indicate that the computational time is relatively higher for smaller sizes of blocks. In contrast to computational time, the SNR and SFR are low for the smaller size of blocks. The desirable values for better fusion are higher SNR and SFR and thus a tradeoff block size ($=8 \times 8$) was chosen for fusion.

In this case, all three images (Landsat-8, LISS-IV, and Sentinel-2A) are passed through a low pass filter to derive an artificial image. The input images, $I(r, c)$, and their respective artificial blurred images were divided into $B \times B$ ($=8 \times 8$) blocks, and then the DCT coefficients were determined for each block. The DCT of a two-dimensional block can be determined using Eqn. (1).

$$d(i, j) = \frac{2\alpha(i)\alpha(j)}{B} \times \sum_{r=0}^{B-1} \sum_{c=0}^{B-1} I(r, c) \cos \left[\frac{(2r+1)\pi k}{2B} \right] \times \cos \left[\frac{(2c+1)\pi k}{2B} \right] \quad (1)$$

In Eqn. (1), $d(0, 0)$ is the directly concatenated (DC) coefficient, which represents the mean value of the image block. All other $d(i, j)$ represents the alternate concatenated (AC) coefficient of the block. $i, j, r, c \in 0, 1, \dots, B-1$,

The value of $\alpha(i)$ can be determined as Eqn. (2).

$$\alpha(i) = \begin{cases} \frac{1}{\sqrt{2}} \text{ if } i = 0 \\ 1 \text{ otherwise.} \end{cases} \quad (2)$$

Thereafter, the correlation coefficients between the original DCT coefficient blocks and the corresponding blurred DCT coefficient blocks were determined to generate the decision map. The correlation coefficient of two $B \times B$ blocks of any two satellite images (Landsat-8, LISS-IV, and Sentinel-2A) can be determined using Eqn. (3).

$$\text{Corr}(I_1, I_2) = \frac{\sum_{r=0}^{B-1} \sum_{c=0}^{B-1} (I_1(r, c) - \bar{I}_1)(I_2(r, c) - \bar{I}_2)}{\sqrt{\sum_{r=0}^{B-1} \sum_{c=0}^{B-1} (I_1(r, c) - \bar{I}_1)^2} \sqrt{\sum_{r=0}^{B-1} \sum_{c=0}^{B-1} (I_2(r, c) - \bar{I}_2)^2}} \quad (3)$$

where $I_1(r, c)$ is the intensity of the $(r, c)^{\text{th}}$ pixel of the image I_1 , $I_2(r, c)$ is the intensity of the $(r, c)^{\text{th}}$ pixel of the image I_2 , $\bar{I}_1$ is the mean intensity value of the image I_1 , and $\bar{I}_2$ is the mean intensity value of the image I_2 .

The correlation coefficient between the two $B \times B$ blocks in the DCT domain of images I_1 and I_2 can be derived using Eqn. (4).

$$\text{Corr}_{DCT}(I_1, I_2) = \frac{\sum_{i=0}^{B-1} \sum_{j=0}^{B-1} P_{I_1}(i, j) \times P_{I_2}(i, j)}{\sqrt{\sum_{i=0}^{B-1} \sum_{j=0}^{B-1} [P_{I_1}(i, j)]^2} \sqrt{\sum_{i=0}^{B-1} \sum_{j=0}^{B-1} [P_{I_2}(i, j)]^2}} \quad (4)$$

Table 2
Acquisition of satellite data and their detailed information.

Sl. No.	Acquisition date	Product ID	Satellite Sensor	Resolution (m)	Wavelength range of bands (μm)	Composite Bands (RGB)
1	20th May 2019	LC08L2SP1400442019052020200828	Landsat-8	30	0.85 – 0.88 0.64 – 0.67 0.53 – 0.59	Band-5 Band-4 Band-3
2	01st May 2019	202,524,411	LISS-IV	5.8	0.77—0.86 0.62—0.68 0.52—0.59	Band-4 Band-3 Band-2
3	08th May 2019	L1CT45QVGA02023620190508T045325	Sentinel-2A	10	0.82—0.86 0.63—0.69 0.52 – 0.59	Band-8 Band-4 Band-3

The value of $P_{I_1}(i, j)$ and $P_{I_2}(i, j)$ can be derived using Eqns. (5) and (6), respectively.

$$P_{I_1}(i, j) = d_{I_1}(i, j) - \bar{d}_{I_1} \quad (5)$$

$$P_{I_2}(i, j) = d_{I_2}(i, j) - \bar{d}_{I_2} \quad (6)$$

where $d_{I_1}(i, j)$ and $d_{I_2}(i, j)$ are the DCT coefficients of the $B \times B$ block of images I_1 and I_2 , respectively. Also, $\bar{d}_{I_1}$ and $\bar{d}_{I_2}$ are the mean values of DCT coefficients of the $B \times B$ block of images I_1 and I_2 , respectively.

The correlation coefficient between the image blocks before and after passing the filter in the proper focussed area is less than the improperly focussed area and thus the decision map was prepared based on the minimum correlation coefficient value. That is, the area that shows significant change after passing the filter has less correlation coefficient between the blocks before and after passing the filter and is thus assigned as a fused image value. The fused map was derived from the correlation coefficient values in the DCT domain using a specific decision rule, as demonstrated in Eqn. (7).

$$\text{Fused map} = \begin{cases} I_1 \text{ if } \text{Corr}_{\text{DCT}}(I_1, \bar{I}_1) < \text{Corr}_{\text{DCT}}(I_2, \bar{I}_2) \\ I_2 \text{ if } \text{Corr}_{\text{DCT}}(I_1, \bar{I}_1) > \text{Corr}_{\text{DCT}}(I_2, \bar{I}_2) \\ \frac{I_1 + I_2}{2} \text{ otherwise} \end{cases} \quad (7)$$

The fused map was examined using a consistency verification value.

After consistency verification, a new fused image was derived by applying the inverse of the DCT transform using Eqn. (8) as follows.

$$I(r, c) = \frac{2}{B} \times \sum_{r=0}^{B-1} \sum_{c=0}^{B-1} \alpha(i) \alpha(j) d(i, j) \cos \left[\frac{(2r+1)\pi k}{2B} \right] \times \cos \left[\frac{(2c+1)\pi k}{2B} \right] \quad (8)$$

In this study, a stepwise fusion of the images was made to fuse the three-sensor data. That is, two sensor data were fused first, and then the fused image was further fused with the third sensor data.

The fused satellite image was further used for the extraction of the training and validation image dataset for DCNN and DNN model design for LULC classifications. For the development of the DCNN model, a large number of image data is needed. In this study, 6000 image samples of 6×6 sizes were extracted from the fused satellite image for five specified LULC classes (barren land, coal mining region, water body, built-up area, and vegetation) to train and validate the DCNN model. That is, 1200 image samples of 6×6 size were derived for each LULC class. The samples for each land-use type were extracted by envisaging the high-resolution Google Earth imagery and past information about the field characteristics (shape, texture, and colour). The characteristics of selected LULC types in the region are defined as follows:

- Barren Land: unproductive soil covered with no kind of green.
- Built-up Area: commercial, residential, and other man-made areas.
- Coal mine region: abandoned or active open pit coal mines.
- Vegetation: forest, grassland, agricultural land, and shrublands.
- Waterbody: pond, river, lakes, and drain.

For pixel-based classification using DNN, all the pixels lay within the 6000 selected image samples were considered as input. In each image, 36 pixels are located, and each pixel consists of three pieces of information. The values of each pixel of the 6000 image samples were stored in the database for DNN model training and validation. Thus, the total number of data extracted was $6000 \times 36 \times 3$ for five classes. That is, the number of samples or input for each class was $1200 \times 36 \times 3$.

2.2.3. Model development and evaluation

Deep learning models are widely used in satellite image classification for object and pixel-based classifications (Storie et al., 2018; Zhao and Du, 2016). The multi-sensor fused image in particular consists of a massive matrix of the spectral value of the pixel. In this study, the DCT-based spatial correlation method was used to retain

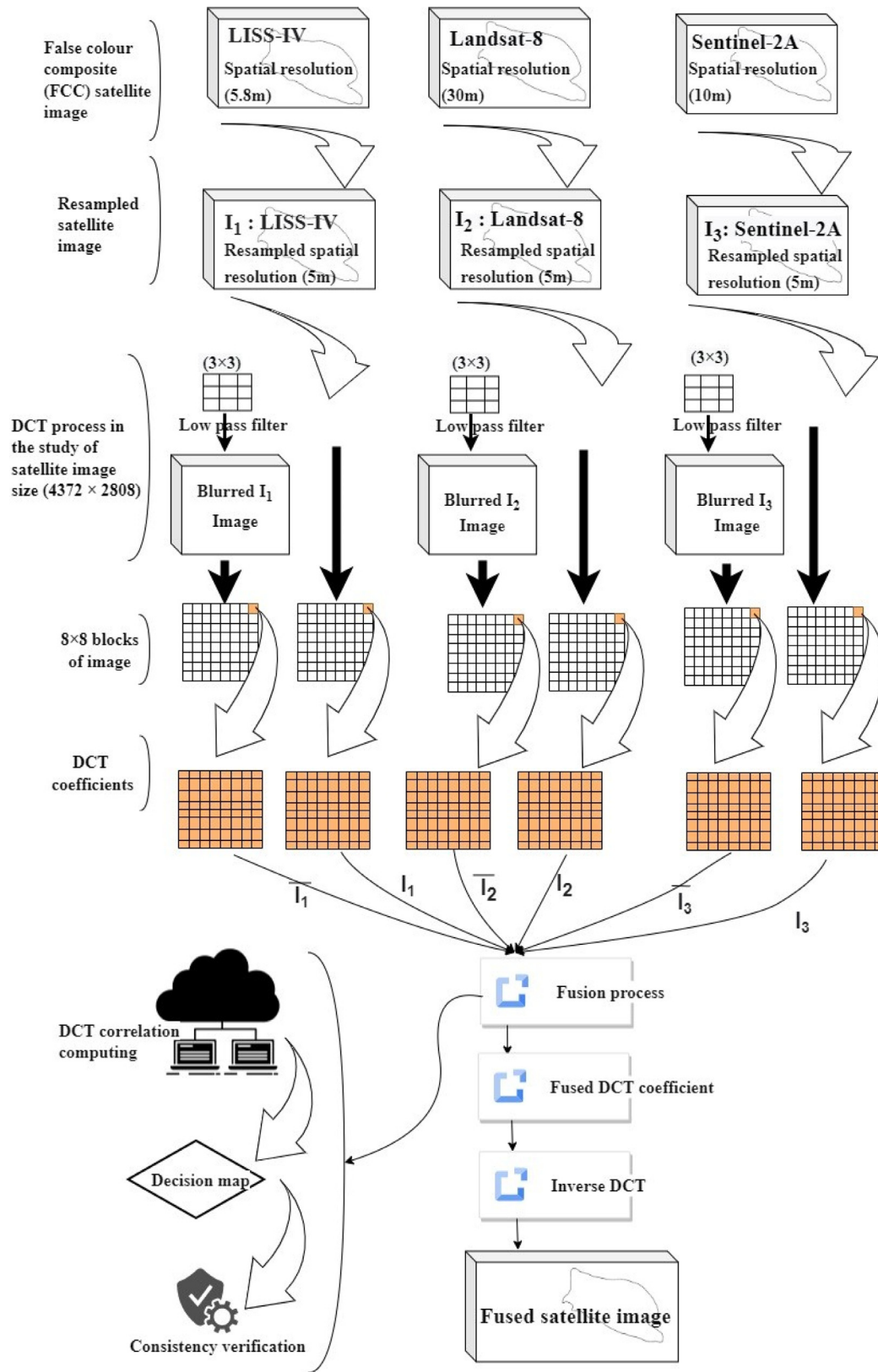


Fig. 3. The DCT framework of spatial correlation coefficient-based satellite image fusion.

the information from that image that best represents the true scene content. Fig. 4 depicts the architectures of DCNN classification and DNN classification models.

DCNN is a typical multi-layer perceptron neural network model in which large numbers of two-dimensional images are used as input for model training to classify into

Table 3
Performance measurement of DCT-based fusion process for different block sizes.

Block size (rows × columns)	DCT_CT (second)	DCT_SNR (dB)	DCT_SFR
2 × 2	1183.64	21.0664	0.4979
4 × 4	323.76	21.0809	0.4979
8 × 8	93.00	21.0665	0.7070
64 × 64	12.76	18.8570	0.7070

Note: DCT_CT = DCT computational time, DCT_SNR = DCT signal-to-noise ratio, DCT_SFR = DCT spatial frequency response.

different classes. The feature maps are automatically extracted from each input image for model training. The current DCNN model is designed with three convolutional layers, two max-pooling layers, and one fully connected layer. The outputs from a layer are subsequently used as input to the next layer. The input neurons are locally linked to the preceding layer's output in every convolutional layer. A rectified linear activation function (ReLU) is used in each convolutional layer to eliminate the superfluous neurons. In the DCNN model, the pooling layer was used with the purpose of reducing the spatial extent of the input data after convolution. In this study, a max-pooling function with a 2×2 filter size and 2-stride is used for extracting the maximum value of pixels during convolution. In general, a pooling layer yields a feature map (f_{map}) of $\frac{H-k+1}{s} \times \frac{W-k+1}{s \times B}$ size from $(H \times W \times B)$ image size. Here, H , W , and B are representing the height, width, and number of bands in the input image. k represents the kernel size and s represents the stride length. The input feature maps are convolved in every layer with a specific kernel size to derive the new feature maps. In this, the convolutional layer extracts pertinent spatial information from the image after passing the filter. A similar procedure is followed in every convolutional layer to derive the desirable feature map, which is further linked to the fully connected layer (Goodfellow et al., 2016). The input image layers are convolved with a defined kernel size to derive a convolved feature map (f_{map}), as represented in Eqn. (9).

$$f_{map}[r, c] = (I_f * k)[r, c] = \sum_i \sum_j k[i, j] I_f[r - i, c - j] \quad (9)$$

Where, r and c represent the row and column of the follow-up matrix, respectively, I_f represents the input image, and $k[i, j]$ represents the kernel of size $i \times j$.

The fully connected layer receives the flattened output of the last pooling layer or convolutional layer. The last layer uses the softmax activation function to derive input class probabilities or classifications. The model used stochastic gradient descent with momentum (SGDM), which offers stable accuracy (Dogo et al., 2018; Postalcolu, 2020; Kumar and Gorai 2022a). All the model hyperparameters such as learning rate (LR), momentum (MO), maximum epoch (ME), minibatch size (MBS), learning rate drop period (LRDP), L2Regularization (L2R), learning rate drop factor (LRDF), validation frequency (VAF), and verbose frequency (VEF) are fixed as suggested in Kumar and Gorai (2022a) for training.

On the other hand, DNNs are improved forms of the conventional artificial neural network (ANN), which consists of multiple layers. In this study, the DNN model was designed to predict the LULC class by analyzing the input pixel characteristics that are extracted from the fused image. The architecture of the DNN model is shown in Fig. 4. DNN models can solve complex problems through a wide range of architectures. A DNN model can be designed in multiple ways, including layering structures, layer types, and connecting patterns of the nodes. Further, the specified classes of input neurons pass into the input layer. The output from the input layer passes into consecutive hidden layers and finally, the output from the last hidden layer passes into the output layer.

2.2.4. Model validation and performance measures

The models were validated using various performance measure indices like accuracy, error rate, recall, and precision. These are important parameters for the evaluation of the classification model (Junker et al. 1999; Chicco and Jurman, 2020). All these indices are commonly derived from confusion matrix parameters, including true-positive (TP), false-positive (FP), true-negative (TN), and false-negative (FN). A TP and TN are the outcomes where the model correctly predicts the positive class and the negative class, respectively. On the other hand, FP and FN are the outcomes where the model incorrectly predicts the positive class and the negative class, respectively. Further, these performance measure indices are calculated using Eqns. (10–13).

$$Accuracy = \frac{TP + TN}{(TP + FP + TN + FN)} \quad (10)$$

$$Error = \frac{FP + FN}{(TP + FP + TN + FN)} \quad (11)$$

$$Precision = \frac{TP}{(TP + FP)} \quad (12)$$

$$Recall = \frac{TP}{(TP + FN)} \quad (13)$$

3. Results

The performance evaluation of the model classifiers was assessed using four performance measure indices (accuracy, error rate, recall, and precision). The performance measure indices were determined separately for the training and val-

validation dataset from the respective confusion matrix parameters (TP, TN, FP, and FN) for both models. These are summarized in Table 4.

The confusion matrix of the DCNN classifier indicates that the TP values for BL, BA, CM, VE, and WB are respectively 840, 835, 839, 840, and 837 for the training dataset, and the respective values for the validation dataset are 360, 354, 355, 360, and 357. Similarly, the TN numbers for BL, BA, CM, VE, and WB are respectively 3358, 3358, 3360, 3359, and 3356 for the training dataset and the respective values for the validation dataset are 1615, 1440, 1437, 1436, and 1434. The FP numbers for BL, BA, CM, VE, and WB are respectively 2, 2, 0, 1, and 4 for the training dataset, and the respective values for the validation dataset are 5, 0, 3, 0, and 6. The FN numbers BL, BA, CM, VE, and WB are respectively 0, 5, 1, 0, and 3 for the training dataset, and the respective values for the validation dataset are 0, 6, 5, 0, and 3.

The confusion matrix of the DNN classifier indicates that the TP values for BL, BA, CM, VE, and WB are respectively 760, 556, 803, 840, and 622, for the training dataset, and the respective values for the validation dataset are 276, 202, 314, 293, and 195. Similarly, the TN numbers for BL, BA, CM, VE, and WB are respectively 3123, 3239, 3179, 3356, and 3279, for the training dataset and the respective values for the validation dataset are 1162, 1183, 1135, 1271, and 1225. The FP numbers for BL, BA, CM, VE, and WB are respectively 252, 131, 173, 4, and 59 for the training dataset, and the respective values for the validation dataset are 106, 59, 90, 1, and 29. The FN numbers BL, BA, CM, VE, and WB are respectively 65, 274, 45, 0, and 235 for the training dataset, and the respective values for the validation dataset are 21, 121, 26, 0, and 117.

The confusion matrix parameters (TP, TN, FP, and FN) were used to measure the model performance indices using Eqns. 10–13. The results are summarized in Fig. 6(a)–(d). Fig. 6(a) indicates that classification accuracies of the DCNN model and DNN model ranged from 99.83% to 99.99% and 90.36% to 99.90% for the training dataset and the same ranged from 99.50% to 99.99% and 88.50% to 99.94% for the validation database for five LULC classes. In the DCNN model, the accuracy level was found to be approximately equal for each class for both the training and validation datasets. On the other hand, the DNN model offers relatively less accuracy for each class in comparison to the DCNN model for both the training and validation datasets. Moreover, the classification accuracy for the built-up area was found to be significantly less in comparison to other classes in the DNN model. This may be due to the existence of minor or temporary construction areas in the region.

Similarly, the results shown in Fig. 6(b) indicate that the error rates of the DCNN model and DNN model ranged from 0.01% to 0.17% and 0.01% to 9.64% on the training dataset for five classes, and the same ranged from 0.01% to 0.50% and 0.06% to 11.50% on the validation dataset.

In the DCNN model, the training dataset showed the lowest error for the vegetation class, and the highest error in classes of built-up area, and waterbody. Also, the validation dataset offers the lowest error for the vegetation class and the highest error for the waterbody. Similarly, in the DNN model, both the training and validation datasets exhibit the lowest error for the vegetation class and the highest errors in the built-up area.

In addition to the accuracy and error rate, two other indices (Precision, and Recall) were used to measure the effectiveness of the models. Fig. 6(c) reveals that classification accuracies of the DCNN model and DNN model ranged from 99.52% to 99.99% and 75.10% to 99.53% for the training dataset and the same ranged from 98.35% to 99.99% and 72.25% to 99.66% for the validation database for five LULC classes. A higher precision score gives a higher TP rate or a low FP rate. In the DCNN model, both the training and validation datasets exhibit the lowest precision for the waterbody class and the highest precision for the vegetation class. Similarly, in the DNN model, both the training and validation datasets exhibit the lowest precision for barren land and the highest precision for vegetation class.

Fig. 6(d) indicates that classification accuracies of the DCNN model and DNN model ranged from 99.40% to 99.99% and 66.99% to 99.99% for the training dataset and the same ranged from 98.33% to 99.99% and 62.50% to 99.99% for the validation dataset for five LULC classes. A higher recall value offers a low FN rate. In the DCNN model, both the training and validation datasets exhibit that the recall values for each LULC class are close to the ideal value. On the other hand, in the DNN model, both the training and validation datasets showed low recall values for the built-up area and waterbody, but satisfactory results were observed for the rest of the classes.

4. Discussion

The study used fused data, derived from three different sensors (LISS IV, Landsat 8, and Sentinel 2A), for DCNN and DNN model development. One of the challenging tasks of the proposed research is the fusion of multi-sensor images with varying spatial resolutions (Landsat-8—30 m, LISS-IV – 5.8 m, and Sentinel-2A-10 m). Landsat image has the lowest spatial resolution among the three sensor images. The objective of fusion is to derive a new image that can provide better spatial information. In this study, image resampling was done by using a bilinear interpolation method to derive uniform spatial resolution (=5 m) images for each sensor. This process enhances the quality of the images. A total of 6000 image samples of 6×6 sizes and 216,000 pixels were used to train and validate the DCNN and DNN model development, respectively. That is, a DCNN model uses the object-based dataset, and a DNN model uses the pixel-based dataset for training and validation. For both models, the entire dataset was

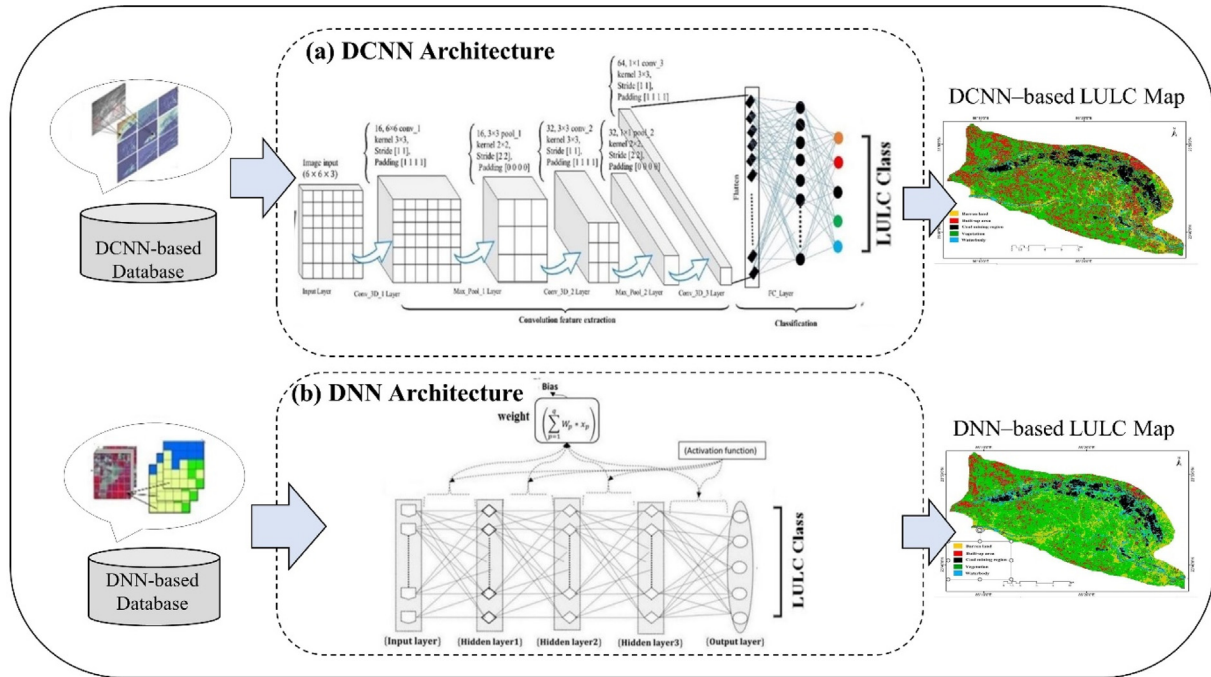


Fig. 4. Framework of DCNN and DNN models.

Table 4

Confusion matrix for training and validation dataset of DCNN classifier and DNN classifier.

Training dataset											
DCNN model						DNN model					
Predicted Class	Actual Class					Predicted Class	Actual Class				
	BL	BA	CM	VE	WB		BL	BA	CM	VE	WB
BL	840	1	0	0	1	BL	760	191	0	0	61
BA	0	835	1	0	1	BA	59	556	18	0	54
CM	0	0	839	0	0	CM	0	55	803	0	118
VE	0	0	0	840	1	VE	2	0	0	840	2
WB	0	4	0	0	837	WB	4	28	27	0	622
Validation dataset											
DCNN model						DNN model					
Predicted Class	Actual Class					Predicted Class	Actual Class				
	BL	BA	CM	VE	WB		BL	BA	CM	VE	WB
BL	360	4	1	0	0	BL	276	90	0	0	16
BA	0	354	0	0	0	BA	20	202	8	0	31
CM	0	0	355	0	3	CM	0	21	314	0	69
VE	0	0	0	360	0	VE	0	0	0	293	1
WB	0	2	4	0	357	WB	1	10	18	0	195

Note: BL = Barren Land, BA = Built-Up Area, CM = Coal Mining Region, VE = Vegetation, and WB = Waterbody.

divided in the ratio of 7:3 for training and validation of the models. For the DCNN model, the values of the hyperparameters were fixed as LR equal to 0.001, LRDF equal to 0.1, LRDP equal to 10, MO equal to 0.95, L2R equal to 0.0001, ME equal to 300, MBS equal to 128, VEF equal to 50, and VAF equal to 30. Kumar and Gorai (2022b) optimized the hyperparameters LR and MO as 0.001 and 0.95, respectively. According to Chee & Li (2020), LR of 0.001 offers the optimum adaptive control to adjust the response of the model to the error estimated with the

update of model weights. Moreover, a DCNN offers good convergence at a 0.95 momentum value (Liu et al., 2020). After fixing the values of the hyperparameters, the progressive losses with iteration were examined for both models. The results revealed that the losses for DCNN and DNN models are 0.2% and 14.7%, respectively, as shown in Fig. 5. The comparative values of the training and validation accuracies of the DCNN model are 99.8% and 99.2%, respectively, and the respective accuracies of the DNN model are 85.3% and 81.8 %, respectively.

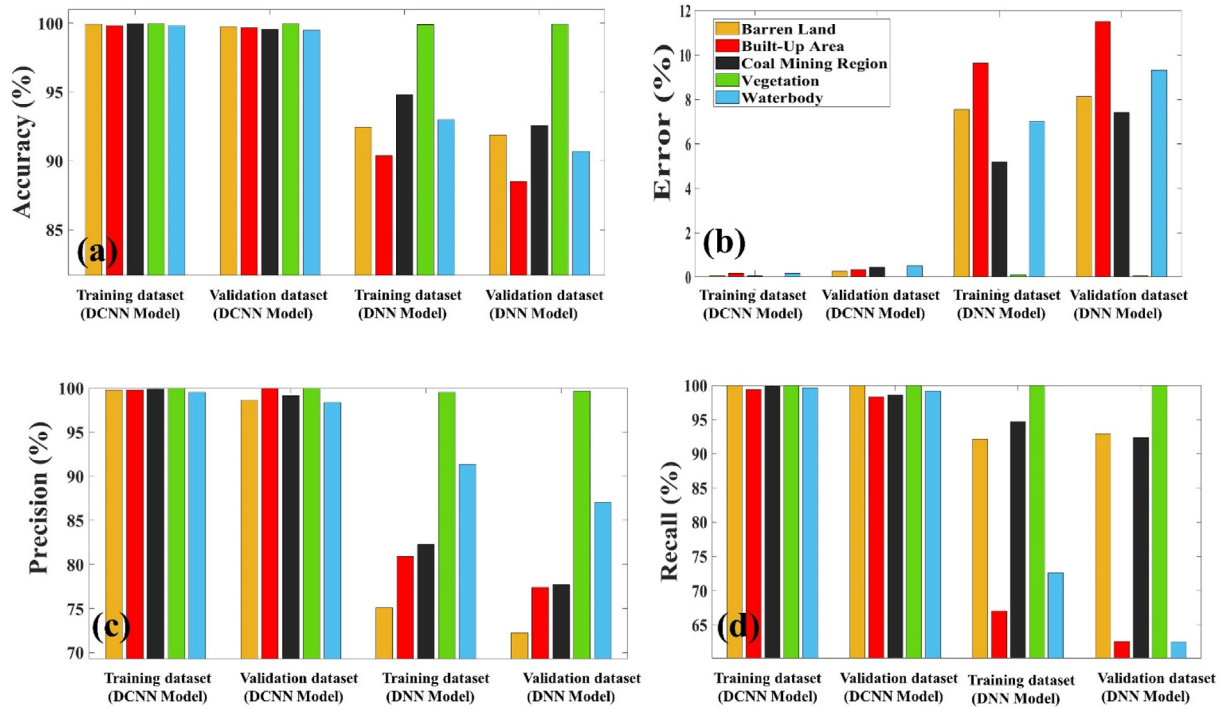


Fig. 5. The progressive loss graph (a) DCNN classification model (b) DNN classification model.

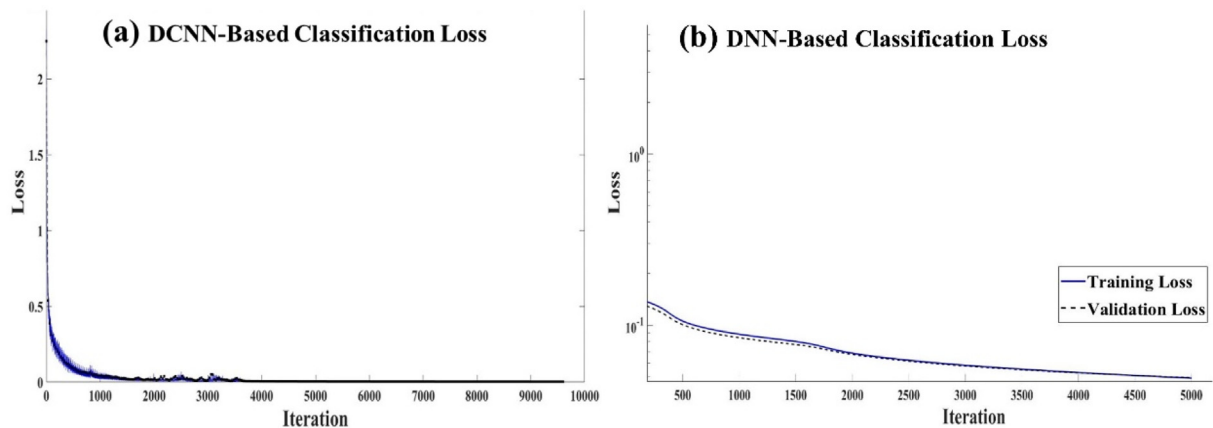


Fig. 6. Graphical representation of the model performance indices of DCNN and DNN classifiers for training and validation datasets (a) accuracy, (b) error rate, (c) precision, and (d) recall.

The classified maps derived using the DCNN model and DNN model are shown in Fig. 7(a) and Fig. 7(b), respectively. Overall, the DCNN model offers better classification accuracies and edge detection in comparison to the DNN model using fused data. The coverage areas of five selected LULC types were determined for both the classified maps that were derived from DCNN and DNN models. The summarized results are represented in Fig. 8. The results indicate that classifications of the regions using the two models showed significant variations. It can be easily inferred that minor built-up areas are classified as either barren land, vegetation, or waterbody in the DNN model. The coal-mining regions are accurately classified in both models. The magnified forms of comparative LULC classi-

fications of two small portions using DCNN and DNN models are shown in Fig. 7. The contemporary high-resolution Google Earth images of the same area are also represented for understanding the delineation accuracy of the two classifiers. The results indicate that the DCNN classifier offers more accurate detection and delineation of pixels in comparison to the DNN classifier. Thus the present investigations revealed that a convolutional-based classifier offers better results than the traditional neural-network-based classifier.

The study also compared the performance of the DCNN model with multi-sensor fused datasets and individual sensor datasets or unfused datasets. The performances of the DCNN model with unfused image datasets, derived from

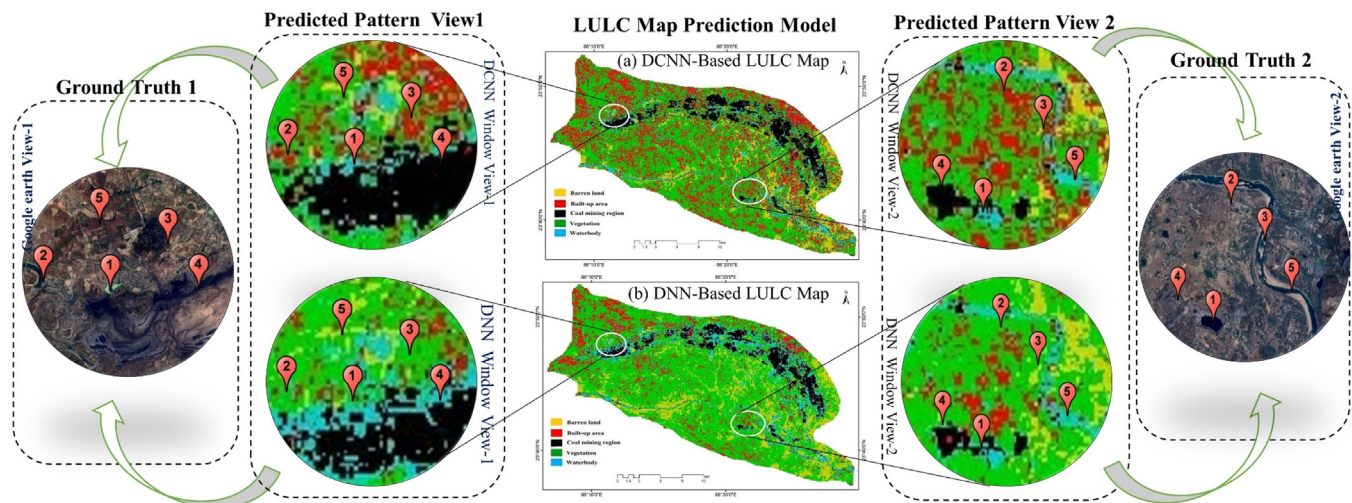


Fig. 7. LULC maps (a) LULC map derived from DCNN classifier (b) LULC map derived from DNN classifier.

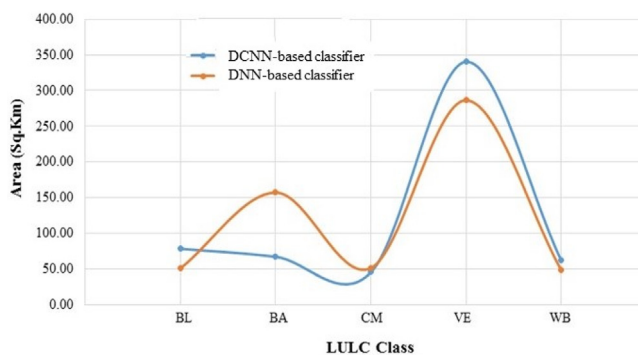


Fig. 8. Trends of area coverages of different LULC classes obtained from the DCNN and DNN-based classifiers.

LISS IV and Landsat, were measured in the studies of Kumar and Gorai (2022a, b). The results indicate that the DCNN model accuracies with the unfused dataset of LISS-IV were found to be 99.04% during training and 87.50% during validation (Kumar and Gorai, 2022b). Similarly, the accuracies with the unfused dataset of Landsat were found to be 95% during training and 88% during validation (Kumar and Gorai, 2022a). Thus, it is clear from the results that the fused image with the DCNN model outperformed (99.8% during training and 99.2% during validation) the unfused image of the same model. The study results exhibit that LULC classification accuracy can be improved by using multi-sensor fused image datasets and thus if resources (multi-sensor data) are available, it is advisable to use the same for obtaining better classification accuracy.

5. Conclusions

The research aim of the current study is to investigate the classification performance of the deep learning algorithms (DCNN and DNN model) using multi-sensor fused datasets for LULC for the mining region. The results indi-

cate that DCNN classification algorithm offers superior performance in terms of the accuracy of LULC classification as compared to the DNN classification algorithms using fused datasets. The DCNN model exhibits higher training and validation accuracies, reaching 99.8% and 99.2% respectively, while the DNN model achieves lower accuracies of 85.3% and 81.8%. Remarkably, the DCNN model offers improved edge detection capabilities as compared to the DNN model, which further enhances the suitability of DCNN for LULC study of mining regions. Overall, the results indicate the superiority of the DCNN model with fused datasets for LULC classification in the mining region over the DNN model.

Furthermore, the study compares the performance of the DCNN model with multi-sensor fused images and individual sensor datasets. The results indicated that the DCNN model with fused data yields higher classification accuracies, achieving 99.8% and 99.2% on the training and validation datasets, respectively. In contrast, the DCNN model using LISS IV data offers training and validation accuracies of 99.04% and 87.5%, respectively, and the same with Landsat data offers training and validation accuracies of 95% and 88%, respectively.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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